

Increased Microerythrocyte Count in Homozygous α (+)-Thalassaemia Contributes to Protection against Severe Malarial Anaemia

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Evidence abstract

BACKGROUND The heritable haemoglobinopathy α (+)-thalassaemia is caused by the reduced synthesis of α -globin chains that form part of normal adult haemoglobin (Hb).

Individuals homozygous for α (+)-thalassaemia have microcytosis and an increased erythrocyte count.

α (+)-thalassaemia homozygosity confers considerable protection against severe malaria, including severe malarial anaemia (SMA) (Hb concentration < 50 g/l), but does not influence parasite count.

We tested the hypothesis that the erythrocyte indices associated with α (+)-thalassaemia homozygosity provide a haematological benefit during acute malaria.

METHODS AND FINDINGS Data from children living on the north coast of Papua New Guinea who had participated in a case-control study of the protection afforded by α (+)-thalassaemia against severe malaria were reanalysed to assess the genotype-specific reduction in erythrocyte count and Hb levels associated with acute malarial disease.

We observed a reduction in median erythrocyte count of approximately $1.5 \times 10^{12}/l$ in all children with acute falciparum malaria relative to values in community children ($p < 0.001$).

We developed a simple mathematical model of the linear relationship between Hb concentration and erythrocyte count.

This model predicted that children homozygous for α (+)-thalassaemia lose less Hb than children of normal genotype for a reduction in erythrocyte count of $>1.1 \times 10^{12}/l$ as a result of the reduced mean cell Hb in homozygous α (+)-thalassaemia.

In addition, children homozygous for α (+)-thalassaemia require a 10% greater reduction in erythrocyte count than children of normal genotype ($p = 0.02$) for Hb concentration to fall to 50 g/l, the cutoff for SMA.

We estimated that the haematological profile in children homozygous for α (+)-thalassaemia reduces the risk of SMA during acute malaria compared to children of normal genotype (relative risk 0.52; 95% confidence interval [CI] 0.24-1.12, $p = 0.09$).

CONCLUSIONS The increased erythrocyte count and microcytosis in children homozygous for α (+)-thalassaemia may contribute substantially to their protection against SMA.

A lower concentration of Hb per erythrocyte and a larger population of erythrocytes may be a biologically advantageous strategy against the significant reduction in erythrocyte count that occurs during acute infection with the malaria parasite *Plasmodium falciparum*.

This haematological profile may reduce the risk of anaemia by other *Plasmodium* species, as well as other causes of anaemia.

Other host polymorphisms that induce an increased erythrocyte count and microcytosis may confer a similar advantage.

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Evidence checklist

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2. Individuals homozygous for alpha(+)-thalassaemia have microcytosis and an increased erythrocyte count.
3. Alpha(+)-thalassaemia homozygosity confers considerable protection against severe malaria, including severe malarial anaemia (SMA) (Hb concentration < 50 g/l), but does not influence parasite count.
4. We tested the hypothesis that the erythrocyte indices associated with alpha(+)-thalassaemia homozygosity provide a haematological benefit during acute malaria.
5. METHODS AND FINDINGS Data from children living on the north coast of Papua New Guinea who had participated in a case-control study of the protection afforded by alpha(+)-thalassaemia against severe malaria were reanalysed to assess the genotype-specific reduction in erythrocyte count and Hb levels associated with acute malarial disease.
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8. This model predicted that children homozygous for alpha(+)-thalassaemia lose less Hb than children of normal genotype for a reduction in erythrocyte count of $>1.1 \times 10^{12}/l$ as a result of the reduced mean cell Hb in homozygous alpha(+)-thalassaemia.
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11. CONCLUSIONS The increased erythrocyte count and microcytosis in children homozygous for alpha(+)-thalassaemia may contribute substantially to their protection against SMA.
12. A lower concentration of Hb per erythrocyte and a larger population of erythrocytes may be a biologically advantageous strategy against the significant reduction in erythrocyte count that occurs during acute infection with the malaria parasite *Plasmodium falciparum*.
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14. Other host polymorphisms that induce an increased erythrocyte count and microcytosis may confer a similar advantage.

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